TECHNICAL NOTICE

Third-party Notices and Runtime Components

MediaPipe provenance, modifications, checksums, licensing, and update procedure.

2026 EDITION / IAIN BENNETT / MIT LICENSE

The **VirtualGlove Controller (Arduino UNO Q)** hosts the camera and recognition runtime described below.

VirtualGlove's original source code and associated documentation are licensed under the repository's MIT License. That license does not replace the licenses or terms that apply to third-party software and model files.

What an ordinary release distributes

Item	Included in release	Licence and notice
VirtualGlove source, documentation, and original artwork	Yes	MIT; see `LICENSE`
MediaPipe 0.10.35 ARM64/Python 3.12 wheel	Yes	Apache 2.0; the wheel retains its own licence and the release includes `licenses/Apache-2.0.txt`
Google Hand Landmarker model	Yes	Apache 2.0; see `licenses/Apache-2.0.txt` and the model record below
VirtualGlove Nestopia patch and reproducible build recipe	Yes	GNU GPL version 2; see `licenses/GPL-2.0.txt`
Compiled `lr-nestopia-powerglove` core	Built on RetroPie when the user selects native support; not bundled in the Controller archive	GNU GPL version 2; the installer preserves upstream copying information with the installed core
`uhubctl`	Installed from Debian only when the camera-recovery option is used; not bundled	GNU GPL version 2 or later, under the Debian package's own notices
RetroArch, FCEUmm, stock Nestopia, and RetroPie	Already supplied by or installed through RetroPie; not bundled	Their respective upstream licences
Arduino platform and libraries listed below	Downloaded by the Arduino toolchain; not bundled in the Controller archive	Their respective upstream licences

Keep `LICENSE`, this notice, `licenses/Apache-2.0.txt`, and `licenses/GPL-2.0.txt` with redistributed copies. The application does not distribute ROM images, original game artwork, or Nintendo software.

MediaPipe 0.10.35 ARM64 wheel

A wheel (`.whl`) is an installable Python package. VirtualGlove ships one compiled Linux ARM64 MediaPipe runtime, so the VirtualGlove Controller does not build MediaPipe during installation. The `cp312-cp312` tags identify CPython 3.12 and its binary interface; `linux_aarch64` identifies ARM64 Linux.

```
python/worker-wheels/mediapipe-0.10.35+powerglove.cpu1-cp312-cp312-linux_aarch64.whl
```

Property	Value
Component	MediaPipe 0.10.35 for CPython 3.12, Linux ARM64
Upstream project	<https://github.com/google-ai-edge/mediapipe>
Upstream source commit	`f8ef212d5c962c0e853db7e59d217056b187084b`
License	Apache License 2.0
VirtualGlove packaged-wheel SHA-256	`6d29bfc33daebd8e47ff9a75d09ae8c032cdcc74445ba365c5aa78a85a6a2d2e`

Modification notice

The ARM64/Python 3.12 wheel was built from the identified upstream source for the UNO Q environment. It retains upstream source headers and MediaPipe's full Apache 2.0 license at `mediapipe-0.10.35+powerglove.cpu1.dist-info/licenses/LICENSE`. The build keeps the established MediaPipe Hands graph used by VirtualGlove and includes the narrow Linux compatibility and GPU-research support recorded in the Engineering Journey. Production selects the four-thread XNNPACK CPU graph; the slower GPU lanes are not selected during gameplay.

The release wheel was then repackaged for the headless Controller. Its `RECORD` integrity list was rebuilt after these metadata changes:

- The unused `jax` dependency declaration was removed.
- The unused `jaxlib` dependency declaration was removed.
- Headless OpenCV was pinned to `opencv-contrib-python-headless==4.11.0.86`.

Removing unused JAX dependencies avoids a large first-start download and reduces pressure on the Controller's storage. The rebuilt wheel was imported on the ARM64 Controller, reported MediaPipe `0.10.35+powerglove.cpu1` and OpenCV `4.11.0`, and confirmed that JAX was absent. Do not substitute another wheel without repeating dependency, camera, recognition, replay, and thermal tests.

Google Hand Landmarker model

VirtualGlove uses Google's float16 Hand Landmarker task bundle.

Property	Value
VirtualGlove Controller runtime path	`data/models/hand_landmarker.task`
Official download	<https://storage.googleapis.com/mediapipe-models/hand_landmarker/hand_landmarker/float16/1/hand_landmarker.task>
SHA-256	`fbc2a30080c3c557093b5ddfc334698132eb341044ccee322ccf8bcf3607cde1`
Size	7,819,105 bytes

The unmodified model is preserved in `models/hand_landmarker.task` and included in the App Lab installation ZIP. Its Apache 2.0 license is in `licenses/Apache-2.0.txt`; this guide records its source, checksum, and licensing evidence. Google's official Hand Landmarker documentation links a model card that explicitly states Apache License, Version 2.0 on page 2. The project's MIT license does not replace that license. No model modifications were made.

When vision is first activated, the application verifies and copies the bundled model into private, persistent `data/models/`. A verified cached copy is reused. A damaged cache is replaced from the bundle; a damaged bundle is rejected. Google's pinned download is used only when no bundled model is available. **Gestures off** does not open or install the model. Wi-Fi deployments preserve `data/` and include the bundled recovery copy. The model therefore does not require internet access in a complete installation package; Python and other first-install dependencies may still require downloads.

`scripts/fetch-runtime-assets.sh` follows the same bundled-first policy and verifies the pinned checksum before installing the private cached copy. `models/SHA256SUMS` records the preserved model's digest. Keep the model, license, notices, and checksum together in backups. Store a copy of the recovery archive on another drive or in your regular off-machine backup; copies on the same Mac do not protect against loss of that Mac.

Verified VirtualGlove Controller sketch toolchain

The September 4, 2026 sketch build and firmware deployment used these pins from `sketch/sketch.yaml`:

- Arduino Zephyr platform **1.0.0**
- Arduino_RouterBridge **0.4.3**
- Arduino_RPClite **0.3.0**
- ArxContainer **0.7.0**
- ArxTypeTraits **0.3.2**
- DebugLog **0.8.4**
- MsgPack **0.4.2**

The build resolved Arduino_LED_Matrix **0.1.3** from that platform. Preserve the dependency pins when synchronizing with App Lab; its shortened generated configuration is not a replacement for the project's complete configuration. Revalidate compilation and device operation before changing a pin.

Ordinary UNO Q release archives redistribute two unmodified artifacts from the Arduino Zephyr platform 1.0.0: the UNO Q loader `zephyr-arduino_uno_q_stm32u585xx.elf` and `variants/arduino_uno_q_stm32u585xx/flash_sketch.cfg`. The latter retains its Arduino copyright and Apache-2.0 SPDX header. The platform source is <https://github.com/arduino/ArduinoCore-zephyr/tree/1.0.0>; its Apache 2.0 license is included as `licenses/Apache-2.0.txt`. The adjacent release manifest records exact sizes and SHA-256 values. VirtualGlove's separately compiled Matrix sketch remains MIT-licensed project code; its linked Arduino libraries retain their upstream licenses.

Redistributed platform artifact	SHA-256
`zephyr-arduino_uno_q_stm32u585xx.elf`	`39d4a4fd47241663323f6e04f94dd8f5a9f9ad6582cf1df37f9709b74026adcd`
`flash_sketch.cfg`	`38706cee1f9ff2e53364a47129d1c1aea9bb9687ed26d7d70b4a9f9bc5bca60c`

Modified Nestopia libretro core

The optional `lr-nestopia-powerglove` core is built from libretro Nestopia revision `5a1cd378cb46ca9ccc2dd6f8b2b6a79ab986052e`. Nestopia identifies its license as the GNU General Public License, version 2. VirtualGlove's MIT license does not replace the license of Nestopia or the resulting modified core. The corresponding licence text is distributed as `licenses/GPL-2.0.txt`.

Property	Value
Component	libretro Nestopia with the VirtualGlove native-state patch
Upstream project	<https://github.com/libretro/nestopia>
Pinned revision	`5a1cd378cb46ca9ccc2dd6f8b2b6a79ab986052e`
Upstream license	GNU General Public License, version 2
Local modification	`native/nestopia-powerglove/nestopia-powerglove.patch`
Patch SHA-256	`1cdde475a3d0da13a51d975c1bcacaedb3b9e1324260a4575e655346fcaa692f`
Modified upstream files	`libretro/libretro.cpp`; `source/core/input/NstInpPowerGlove.cpp`
Modification ledger	This guide, under **Nestopia modification ledger**
Build recipe	`scripts/build-nestopia-powerglove.sh`
Built core name	`nestopia_powerglove_libretro.so` on RetroPie
Installed core directory	`/opt/retropie/libretrocores/lr-nestopia-powerglove/`

Ordinary releases contain the patch and build recipe, not a compiled core. If the user accepts the RetroPie installer's optional native-core step, the target machine downloads the pinned upstream source, including its author notices and `COPYING` file, applies the patch, and builds for its own processor. The core installer places `COPYING` and this consolidated VirtualGlove notice and modification ledger beside the installed binary. The original Nestopia copyright/GPL header in `NstInpPowerGlove.cpp` remains byte-for-byte intact, and the build stops if a future patch changes it. Stock Nestopia remains untouched and FCEUmm remains available.

If prebuilt cores are published in the future, produce separately identified artifacts for every tested RetroPie architecture. Accompany each binary with the exact complete corresponding source archive used to build it, the local patch and build instructions, all upstream notices, and the GPLv2 license. A Git commit or patch URL alone is not the project's binary-distribution plan. ROM images are never part of a source or binary core artifact.

At runtime, RetroArch loads the custom core only for an explicitly selected ROM. The launch entry passes the read-only latest-sample file through `POWERGLOVE_NATIVE_STATE`; the default path is `/run/powerglove/native-state`. The patch registers a separately named **VirtualGlove** controller and identifies the library as **Nestopia PowerGlove**. Invalid, stale, uncalibrated, lost-tracking, or wrong-profile samples are neutralized. The compatibility record in <u>Super Glove Ball native compatibility</u> separates exact-ROM-confirmed X/Y/Z and hand-pose packet behavior from wrist and button fields that remain unmapped.

The local patch changes only `libretro/libretro.cpp` and `source/core/input/NstInpPowerGlove.cpp`. SHA-256 values for both pristine upstream files are recorded below. The build applies the patch only after checking out the pinned commit, rejects unrelated source-tree changes, and verifies that Nestopia's original 22-line Power Glove copyright and GPL header remains byte-for-byte identical.

Nestopia modification ledger

This additive record does not replace Nestopia's source-file headers, copyright notices, Git history, or `COPYING` file. At the pinned revision, `NstInpPowerGlove.cpp` begins with Martin Freij's original 2003-2008 copyright and GPL notice. The patch begins after that complete header and leaves it byte-for-byte unchanged. The protected pristine files have these SHA-256 values:

- `libretro/libretro.cpp`: `3e6356e56d1c25e2266be155ea2a0a3155d971e60eecba946026422957f5a25f`
- `source/core/input/NstInpPowerGlove.cpp`: `7328ab1cc9cc902ac129d217c4d47faa247d2d6e7d41b8d8ac7eb634c097e7cd`

The September 4 `libretro.cpp` changes added the versioned latest-sample structure; coherence, profile, calibration, detection, and freshness checks; neutral invalid-input behavior; a separately selectable controller; calibrated X/Y plus Start and Select delivery; the `Nestopia PowerGlove` identity; and callback cleanup. The matching `NstInpPowerGlove.cpp` changes enable native input only when `POWERGLOVE_NATIVE_STATE` is present, provide the exact-ROM ten-byte stream while retaining Nestopia's ordinary twelve-byte path, and add opt-in trace hooks without altering normal latch processing. Cabinet testing corrected camera-to-Nestopia Y orientation; unknown fields remained neutral and FCEUmm remained available.

The September 5 changes added closed-hand and index-point flags without changing the version-1 record size. A five-finger fist maps to packet byte 5 `$FF`, index point to `$0F`, and open or ambiguous poses to `$00`. Calibrated depth maps to absolute signed packet byte 3 with the camera-facing sign reversed. The exact ROM repeatedly received `$00` open, `$FF` fist, `$0F` index point, and fist plus forward-Z (`$81`) Power Punch candidates; lost or stale tracking remained fully neutral.

A September 6 diagnostic option inserts checked hooks after the normal patch. Project-owned `diagnostic_trace.h` records callback validity, sequence and publication identities, and local consumption time, then exports a private CSV on game unload. It does not modify the production patch, upstream headers, native-state ABI, or normal selection and is active only for an explicitly diagnostic build and trace environment.

VirtualGlove 0.4.0 requires matching signed-controller software on both computers, but its coordinate-efficiency work does not change this patch or require a rebuild. The confirmed ten-byte packet is documented in [Super Glove Ball native compatibility]. Bytes 7-8 retain Nestopia's fixed `$00` initialization; their gameplay role is not established.

Future changes must remain in the local patch and be appended here. Do not replace or prepend project ownership over an upstream header. Keep upstream notices and licenses with source and binary distributions. The guarded build stops if the protected Nestopia Power Glove header changes.

uhubctl

The VirtualGlove Controller host installer uses the distribution-provided `uhubctl` command to detect and operate genuine USB per-port power switching. The helper never bundles or modifies this utility, and it never forces it to operate on a hub that it does not report as supported.

Property	Value
Component	`uhubctl` USB hub per-port power control utility
Upstream project	<https://github.com/mvp/uhubctl>
License	GNU General Public License, version 2
Installed by	Debian package manager on the VirtualGlove Controller host

Property	Value
Tested repository candidate	Debian 13 ARM64 `uhubctl` 2.6.0-1
Distribution boundary	Not copied into VirtualGlove source or release archives

The root helper first asks `uhubctl` about the exact allowlisted hub location and camera port without using its force option. Only a positive capability result permits a port cycle. Unsupported hardware retains the project's identity-checked whole-hub driver fallback. Debian remains responsible for the installed binary and accompanying copyright and license files.

External RetroPie emulator dependencies

VirtualGlove uses RetroPie-provided emulator software but does not include those binaries in its installation archives. When either dependency is absent, the RetroPie installer can ask the user's existing RetroPie Setup installation to install it. That operation remains governed by RetroPie and the upstream licenses.

Component	VirtualGlove use	Upstream and license	Distribution boundary
RetroArch	Libretro frontend used to load FCEUmm and `lr-nestopia-powerglove`	RetroArch, GPLv3	Installed by RetroPie; not modified or redistributed by VirtualGlove
FCEUmm	Default NES core for standard D-pad/button mappings and the complete Super Glove Ball fallback	FCEUmm, GPLv2	Stock RetroPie core; not modified or redistributed by VirtualGlove

The deterministic direction benchmark separately builds stock FCEUmm revision `236ccdfc911e84c60fea6b9d0699c2d440a8de14` in an isolated working directory. That pin makes the benchmark reproducible; it does not replace the user's RetroPie core, install FCEUmm, or make the benchmark binary a release artifact.

Updating runtime components

MediaPipe wheel or Hand Landmarker model

Before publishing a wheel or model update, complete these steps. The command reference explains the build and verification scripts.

1. Record the official source URL, version, license, size, and SHA-256 here.
2. Update the pinned values in `src/powerglove_vision/runtime_assets.py`, `scripts/fetch-runtime-assets.sh`, `scripts/verify-app-lab-package.py`, and `models/SHA256SUMS` when changing the model.
3. If repackaging another wheel, record every difference from upstream and retain its license files.
4. Build the App Lab installation ZIP and confirm it contains one wheel, the verified model, its license and notices, and only the root `sketch/` application sketch.
5. Test first-launch offline model installation, download fallback, and checksum verification, background preloading with capture off, first activation after reboot, camera initialization, tracking, the Glove Academy and Dashboard pages, and controller output on the VirtualGlove Controller before publishing the package.

Modified Nestopia core

Before changing the Nestopia revision or native patch:

1. Select an exact upstream commit from the official libretro Nestopia repository. Record the commit, upstream license, affected pristine-file SHA-256 values, and new patch SHA-256 in this document.
2. Update the identical revision pin in `scripts/build-nestopia-powerglove.sh`, this modification ledger, tests, and compatibility/benchmark documents. Do not use a moving branch or tag as the build identity.
3. Rebase `native/nestopia-powerglove/nestopia-powerglove.patch` onto a clean checkout. Preserve all upstream headers and notices. The guarded build must still reject changes to the original `NstInpPowerGlove.cpp` header.
4. Run the native-core, state-bridge, installer, selection, exact-ROM trace, safe-neutral, and direction-response tests. Reconfirm packet length, detection, bit order, boundaries, timing, X/Y/Z orientation, open/fist/index values, Start behavior, tracking-loss release, and the explicit FCEUmm rollback on the cabinet.
5. Build the RetroPie installation archive and verify it contains the patch, build/install recipes, this modification ledger, and upstream notices, but no ROM or compiled core. If publishing a binary separately, provide the exact complete corresponding source and GPL materials described above.

When the benchmark FCEUmm pin changes, record the new official revision in the benchmark document and rerun both the native and standard-joypad lanes. Normal RetroPie FCEUmm and RetroArch upgrades remain the responsibility of RetroPie; retest controller selection and fallback gameplay before claiming compatibility.

Documentation and website assets

The following assets support the guides and browser interface. Their origins are recorded separately from the software, model, and firmware dependencies above.

Documentation illustration provenance

The gesture sheets under `docs/images/gestures/` were generated on September 3, 2026 with OpenAI's image-generation tool from project-authored prompts, then selected and arranged for the VirtualGlove gameplay guide. They are documentation assets, not runtime dependencies. No game screenshots, scans, box art, characters, publisher logos, or other source images were supplied to the generator.

The individual gestures and original Pixel Pal mascot under `docs/images/gestures/v2/` were generated on September 4, 2026 with the same built-in tool, using the project's generated contact sheet as a style reference. Their prompts are preserved in `docs/images/gestures/v2/prompts.json`. The earlier illustrations remain available in their original locations.

The index-curl illustration was subsequently redrawn using a user-supplied hand photograph as its pose reference. Only the illustrated glove is included; the reference photograph is not distributed with the project.

Application screenshots

All application screenshots in `docs/images/` were refreshed from the current VirtualGlove source on September 6, 2026. They cover Dashboard, local Rock Paper Scissors, Glove Academy, personalization, players and hand-setup restoration, Setup and guided pairing, Games, and the Help library. `scripts/capture-guide-screenshots.py` renders the real page templates in an isolated browser with temporary player state and sample telemetry. Camera areas use an explicit "Camera preview omitted" placeholder; no live camera, cabinet, or personal settings are accessed. The script also invokes `tests/browser_setup_pairing.py --screenshots` for pairing states using non-secret fixtures. These captures document the interface, not live delivery or hardware verification. Gesture illustrations and physical matrix photographs remain unchanged. No runtime dependencies are added.

Website icon

The website icon in `assets/virtualglove-icon.png` was derived from the project's `assets/virtualglove-logo.png` on September 6, 2026 using OpenAI's image-generation tool. It isolates the hand-and-target emblem without the wordmark. Browser-tab and Apple touch icon variants were resized from that square artwork. `assets/favicon.ico` contains 16, 32, and 48 pixel variants and is the single browser-tab icon declared by the shared page template. The Apple touch icon is separate. Setup and Help use the same root-relative icon URL; there is no Setup-specific asset directory. Setup is served directly at `/setup`, without a query revision or redirect. Older query-string bookmarks still work. The original logo remains unchanged.